

BRENDA ANAGUE

*AIMS Research and Innovation
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Education

AIMS RIC (Rwanda) & University of KwaZulu-Natal (South Africa)

PhD, in Applied Mathematics

2023-Currently

- **Thesis:** “Physics-Informed Neural Networks Beyond Forward Problems: Source Inversion, Equation Learning, and Convergence Theory”
- **Supervisors:** Dr. Bamdad Hosseini (University of Washington, USA), Dr Issa Karambal (AIMS RIC, Rwanda), Dr Jean Medard Ngnotchouye (University of Kwazulu-Natal, SA).

African Institute for Mathematical Sciences (AIMS)

Limbe, Cameroon

M.Sc in Mathematical Sciences, **Speciality: Data Science**

2021–2022

- **Thesis:** “The Study of Over-parameterization for Physics-Informed Neural Networks”
- **Supervisor:** Dr Yunan Yang (Cornell University, USA).

University of Dschang (UDs)

Dschang, Cameroon

M.Sc in Applied Mathematics

2019–2021

- **Thesis:** “Valuation and Hedging of Options in Discrete Markets”
- **Supervisor:** Pr. Gabriel Deugoué (UDs, Cameroon).

University of Dschang (UDs)

Dschang, Cameroon

B.Sc in Mathematics and Computer Science

2016–2019

- **Thesis:** “Mathematical Modelling of Complex Biological Systems”
- **Supervisor:** Dr. Jean Baptiste Patenou (UDs, Cameroon).

Research Interest

- Deep Learning, Physics Informed Machine Learning, optimization.
- Partial Differential Equations (PDEs).
- Numerical analysis for PDEs.
- Inverse problems and convergence analysis of PINNs.
- Conditional stability based on Carleman estimates for source inversion problems of PDES.
- Equation learning of PDEs.
- Numerical solution of PDEs using Gaussian processes.

Scholarships, Awards and Prizes

- **Poster Prize Deep Learning Indaba (Rwanda).** Aug 2025
Poster prize, awarded by Nvidia.
- **Young African AI Research (YAAIR) Prize Winner, at DL Indaba (Senegal).** Sep 2024

6 months internship awarded by InstaDeep.

- **Poster Prize Deep Learning Indaba (Senegal).** Sep 2024
Women In Machine Learning(WIML) Mentorship, fully sponsored by CISPA.
- **Third place at the Senegalese Serpent Venom Sequence Hackathon at DL Indaba (Senegal).** Sep 2024
1000 usd awarded by InstaDeep.
- **PhD fellowship: DeepMind & AIMS RIC (Rwanda).** Nov 2022–Nov 2026
Fully funded.
- **Scholarship: African Institute for Mathematical Sciences (AIMS) Cameroon.** Sep 2021–Jun 2022
Fully funded scholarship.
- **President’s excellence prize for graduate and postgraduates students (Cameroon).** 2019-2022

Publications, Preprints and Ongoing work

- [1] B. Anague et al., “Convergence analysis of PINNs in solving parabolic PDEs under spectral Barron space regularity. (Ongoing work)”, 2026.
- [2] B. Anague et al., “Equation Learning, Source function and Initial Condition Identification of parabolic PDEs from sparse measurements data. (Ongoing work)”, 2026.
- [3] B. Anague et al., “Physics-Informed Neural Networks for Joint Source and Parameters Estimation in Advection Diffusion Equations”, in *Submitted in Journal of Computers and Mathematics with Applications*, 2025, Preprint <https://arxiv.org/abs/2512.07755>.

Talks and Academic Events Attended

- **Deep Learning Indaba Conference (In Person).** 17–22 Aug 2025
Kigali, Rwanda.
- **Workshop on TorchPhysics: Deep Learning for PDEs with TorchPhysics (In person).** 14–15 Nov 2024
Werner-von-Siemens Centre Berlin, Germany.
- **Deep Learning Indaba Conference (In Person).** 01–08 Sep 2024
Dakar, Senegal.
- **Data Science Africa Summer School and Workshop (In Person).** 03–07 June 2024
Nyeri, Kenya.
- **Workshop: From Theory to Practice in Data Science (In Person).** 15–19 April 2024
Talk on Physics-Informed Neural networks for Atmospheric Inverse Problems AIMS RIC, Kigali, Rwanda.
- **Workshop on AI in Science (In Person).** 05–07 Feb 2024
Imperial College London and AIMS RIC in Kigali, Rwanda.
- **Global Fellow Program 'African Data Science Solutions' (In Person).** 20–24 Nov 2023
Imperial College London and AIMS RIC in Kigali, Rwanda.
- **Deep Learning Indaba Conference (In Person).** 01–10 Sep 2023
Accra, Ghana.
- **International Conference on Learning Representations (In Person).** 01–05 May 2023
Kigali, Rwanda.
- **Workshop on Physics for Machine Learning (In person).** 04–05 May 2023
Kigali, Rwanda.
- **Workshop: From Theory to Practice in Data Science (In Person).** 23–27 Apr 2023

Quantum Leap Africa, Kigali, Rwanda.

- **Doctoral Training School in Data Science (In Person).**

March–April 2023

Quantum Leap Africa, Kigali, Rwanda.

Research Visit

- **Center for Industrial Mathematics (ZeTeM)**

Nov 2024

*Visit to the Technomathematical working group, led by Prof. Dr. Dr. h.c. Peter Maaß.
University of Bremen, Germany.*

Prior Research Experience

- **Internship project (InstaDeep)**

Dec 2024–Apr 2025

Retrieval of biological and chemical crop traits from hyperspectral and multispectral data using PINNs.
Supervisor: Ibrahim Salihu Yusuf, InstaDeep, UK.

- **Master's Thesis (AIMS Cameroon)**

Apr–Jun 2022

The understanding of the training dynamic of PINNs under an over-parameterized regime.
Supervisor: Dr Yunan Yang, Cornell University, USA.

- **First Master's Thesis (University of Dschang)**

Mar–Jul 2021

Build an investment strategy that allows a seller and a buyer to hedge against financial risks.
This project relied on the used of Cox-Ross Rubinstein model and stochastic differential equations such as Black Scholes model.
Supervisor: Prof. Gabriel Deugoué, University of Dschang, Cameroon.

- **Project on Finite Element Methods and Advanced Simulation**

Oct 2020–Jan 2021

Finite Element Methods for Elliptic Problems and Simulation.
Supervisor: Dr. Laurent Tchoualag, University of Dschang, Cameroon.

- **Project on Partial Differential Equations**

Oct 2020–Jan 2021

Hamilton-Jacobi Equation.
Supervisor: Prof. Jean Louis Woukeng, University of Dschang, Cameroon.

- **Project on Variational Methods and Optimization**

Oct 2019–Jan 2020

Sobolev Spaces and the Variational Formulation of Elliptic Boundary Value Problems in N Dimensions.
Supervisor: Prof. Calvin Tadmon, University of Dschang, Cameroon.

- **Bachelor's Thesis**

Mar 2019–Jul 2019

Use of kinetic theory to built a biological model.
Supervisor: Dr. Jean Baptiste Patenou, University of Dschang, Cameroon.

Technical Skills

- **Programming languages:** Python, Jax, Matlab, C++, R.
- **Numerical analysis of PDEs.**
- **PDE analysis:** Banach spaces, Barron spaces, Sobolev spaces, PDE regularity theory and functional analysis.
- **Gaussian processes for solving PDEs.**
- **Inverse problems.**

Industry Experience

Internship:

InstaDeep

Kigali, Rwanda

01 Dec 2024–30 Apr 2025

Internship:

International Bank of Cameroon for Savings and Credit (BICEC)

Douala, Cameroon

Aug 2019–Sep 2019

Languages

- **French:** Native
- **English:** Fluent

REFEREES

- **Dr. Bamdad Hosseini**
Department of Applied Mathematics at University of Washington, USA
Email: bamdadh@uw.edu
- **Dr. Issa Karambal**
Senior Researcher for Research and Development and Junior Research Chair in Data Science, AIMS RIC Rwanda
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- **Dr. Jean Medard Ngnotchouye**
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